

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Endowing large language models (LLMs) with the capacity to actively interrogate external knowledge sources constitutes an indispensable prerequisite for addressing complex, real-world tasks. Prevailing approaches, however, remain fundamentally circumscribed: they either depend upon prompting to elicit the model’s latent agentic faculties, or, when reinforcement learning (RL) is brought to bear on intricate interactive settings, they succumb to performance ceilings and training collapse, thereby leaving the model’s genuine agentic potential largely unrealized. To surmount these limitations, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, a refined RL algorithm engineered for robust agent training through the conjunction of sequence-level optimization and dynamic sample filtering. Our model is trained exclusively via RL to interleave multi-turn search with reasoning, thereby obviating any reliance on supervised demonstration data. Across a diverse array of QA benchmarks, our 7B model attains a **34.1%** improvement over a comparable prior work and, more strikingly, surpasses the 14B model of that prior work on complex multi-hop QA such as HotpotQA by nearly **9% relative**, all the while preserving exceptional training stability.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have exhibited remarkable proficiency across a broad spectrum of specialized domains, encompassing mathematical reasoning (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code synthesis (Zheng et al., 2023; Yang et al., 2024), and creative composition (Chakrabarty et al., 2024; Marco et al., 2024). A fundamental limitation nonetheless persists: the knowledge encoded within these models is inherently static, irrevocably confined to the corpus upon which they were trained. To transcend this knowledge cutoff, a now-dominant paradigm endows LLMs with search capabilities, thereby transfiguring them into agents that can actively interrogate external knowledge repositories (Jin et al., 2025b). Such a capability constitutes a paradigmatic instance of tool-calling (Schick et al., 2023), wherein the model learns to orchestrate its interaction with an external search tool in order to resolve problems that lie beyond its unaided reach. The acquisition of this competence necessitates the learning of an intricate, multi-step policy, a formulation that casts the task as a sequential decision-making problem ideally suited to Reinforcement Learning (RL).

In contradistinction to Supervised Fine-Tuning (SFT), which is predicated upon costly static demonstrations and remains constitutionally incapable of instilling exploratory behavior, RL furnishes a principled framework through which LLMs may acquire effective policies via trial-and-error (Chu et al., 2025). In consequence, value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have ascended to the status of a dominant paradigm, esteemed for their conceptual simplicity and their attenuated memory footprint. Notwithstanding its demonstrable success in more constrained regimes, the application of GRPO to the open-ended domain of interactive search exposes critical instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). This fragility is traceable to two foundational deficiencies. First, as delineated by Zheng et al. (2025), GRPO’s token-level objective is ill-posed once it is coupled with a sequence-level reward, an incongruity that engenders high-variance gradients and thereby destabilizes training. Second, the reward sparsity intrinsic to search tasks frequently yields sample groups whose outcomes are homogeneous—uniformly successful or uniformly unsuccessful—whereupon the advantage signal collapses and no learning signal remains, a pathology that severely curtails sample efficiency (Yu et al., 2025; Liu et al., 2025).

To confront these core challenges of instability and inefficient learning in concert, we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an algorithm that synthesizes and refines several salient principles drawn from recent policy optimization research. DSPO appropriates the sequence-level optimization of GSPO (Zheng et al., 2025) so as to reconcile the unit of optimization with the unit of reward; this alignment of the optimization objective with the reward signal fundamentally stabilizes the learning process for long-horizon reasoning tasks. Beyond this,

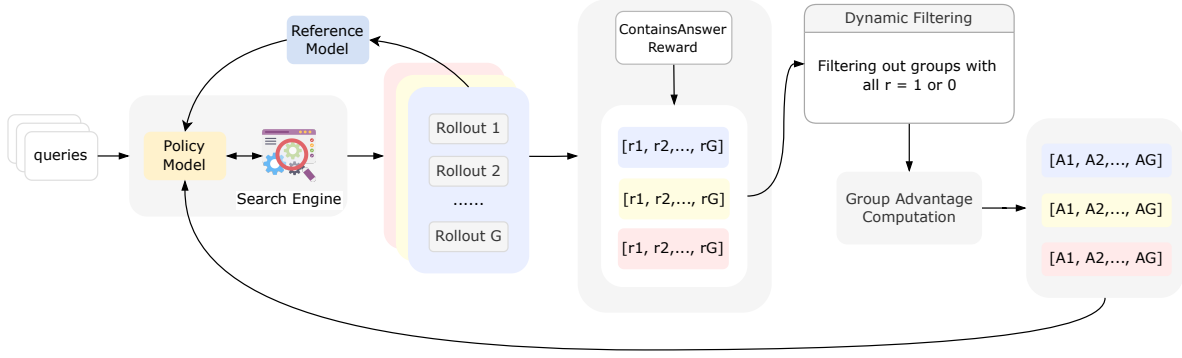


Figure 1: An overview of the DSPO training loop. For a given query, the policy model generates a group of G trajectories through its interaction with the search environment, whereupon each trajectory is assigned a sparse terminal reward. The **dynamic filter** thereupon discards those groups whose outcomes are homogeneous and continues sampling until a batch is populated, thereby ensuring that every training batch furnishes an effective advantage signal. Advantages are subsequently computed and employed to update the policy model by means of a sequence-level objective.

DSPO assimilates a dynamic outcome-based filtering mechanism inspired by DAPO (Yu et al., 2025), a component that actively constructs training batches from rollout groups exhibiting both successful and unsuccessful outcomes for each prompt, thereby guaranteeing that the advantage signal $\hat{A}_i$ remains effective and stable. Through the integration of these two constituents into a single, internally coherent framework, DSPO furnishes a stable and high-performance algorithm expressly tailored to complex, multi-turn search and reasoning tasks. The resulting model attains a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b) and, more remarkably still, surpasses its 14B counterpart (Jin et al., 2025a) on complex multi-hop benchmarks such as HotpotQA, outperforming it by **nearly 9% relative** (0.613 vs. 0.563).

In summary, the principal contributions of this work are the following:

- We propose DSPO, a refined RL algorithm that surmounts the core instability and sample-inefficiency afflictions endemic to the training of agentic search models. It accomplishes this by unifying two key principles within a single cohesive framework: **sequence-level optimization**, which underwrites robust policy updates, and **dynamic outcome-based filtering**, which secures a dense and effective learning signal.
- We substantiate DSPO’s considerable performance gains through rigorous benchmarking. Our 7B model realizes a **34.1% relative improvement** over a comparable 7B baseline and, more strikingly, outperforms its 14B counterpart on complex multi-hop QA, achieving a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563).
- We adduce extensive empirical evidence for DSPO’s superior training stability, demonstrating that it engenders a stable learning trajectory. Crucially, these results are obtained using nothing more than a rudimentary BM25 retriever, a choice that isolates the observed performance gains to the intrinsic robustness of our algorithm.

2 RELATED WORK

2.1 RL FOR LLMs

The landscape of RL for LLMs has undergone rapid evolution, progressing from the foundational Reinforcement Learning from Human Feedback (RLHF) methods that rely on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler, direct-optimization frameworks such as DPO (Rafailov et al., 2023). A pivotal reorientation toward value-free optimization is inaugurated by Group Relative Policy Optimization (GRPO) (Shao et al., 2024), which streamlines training by deriving its reward signal from group-level statistics. GRPO’s token-level objective is nevertheless known to precipitate training instability (Liu et al., 2025; Cui et al., 2025), an infirmity that has prompted a succession of targeted refinements. GSPO redresses this deficiency by transpositioning to a sequence-level objective that matches the unit of reward (Zheng et al., 2025), whereas DAPO confronts

the inefficiency of learning from sparse rewards by means of a dynamic outcome-based sampling mechanism (Yu et al., 2025). In a cognate vein, GMPO stabilizes the token-level objective through a geometric-mean aggregation that attenuates sensitivity to outliers (Zhao et al., 2025). Notwithstanding these advances, we observed in our experiments that these algorithms continue to be beset by challenges such as training collapse and performance bottlenecks. Building upon the aforementioned body of research, we propose our refined algorithm, which synthesizes the principles of sequence-level optimization and dynamic filtering within a unified framework so as to overcome the distinctive challenges posed by the training of autonomous search agents.

2.2 LLMs WITH AGENTIC RETRIEVAL

In order to ameliorate the static-knowledge limitations of LLMs, RAG incorporates external retrievers that dynamically assimilate evolving information (Lewis et al., 2020; Gao et al., 2023). Classical RAG frameworks marshal dense retrievers to procure relevant documents, which are thereafter concatenated into the LLM’s input for generation (Karpukhin et al., 2020). Such approaches, however, are frequently predicated upon fixed pipelines, a rigidity that circumscribes autonomy in complex, multi-turn scenarios. More recently, research has gravitated toward agentic paradigms in which LLMs operate as autonomous agents capable of planning, searching, and reasoning iteratively. Frameworks such as ReAct synergize reasoning with acting, thereby enabling LLMs to interact with tools for tasks such as web navigation (Yao et al., 2023), while multi-agent systems, including AutoGen, orchestrate collaborative workflows (Wu et al., 2024). Recent innovations place particular emphasis on the integration of agentic RAG with RL, whereby agents augment retrieval through deliberate decision-making. Wu et al. (2025) introduce Agentic Reasoning, a framework that integrates external tools to streamline LLM reasoning. Several RL-integrated approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs to interleave reasoning and search through RL alone. This end-to-end paradigm internalizes agentic capabilities and thereby obviates the engineering overhead attendant upon multi-agent frameworks. Yet these methods continue to grapple with training instability and with performance limitations peculiar to the open-ended search domain. Our work confronts these bottlenecks directly: DSPO provides a robust and efficient training framework that guarantees stable policy optimization, thereby enabling LLMs to acquire effective multi-turn search strategies.

3 METHODOLOGY

In this section, we first formulate the task of agentic search as an RL problem and survey prior policy optimization algorithms, foregrounding the limitations they exhibit in this context. We thereupon introduce our proposed algorithm, **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, expounding the core components that underpin its training stability and training efficiency. Finally, we present the integrated training procedure in its entirety.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. The training of LLMs via RL commonly enlists policy gradient methods such as PPO (Schulman et al., 2017), an algorithm widely favored for LLM alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective evaluated on samples drawn from an old policy $\pi_{\theta_{\text{old}}}$. The objective, averaged over tokens, assumes the following form:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO, however, is contingent upon a separately trained value model that estimates token-level advantages $\hat{A}_t$ through Generalized Advantage Estimation (GAE) (Schulman et al., 2015), a dependency that introduces substantial memory overhead and can itself become a wellspring of instability.

To circumvent this, GRPO (Shao et al., 2024) was proposed. GRPO dispenses with the value model altogether by sampling a group of G responses $\{y_i\}_{i=1}^G$ for a given prompt x , and thereafter computes the advantage of each response by normalizing its reward against the aggregate statistics of the group. As with PPO, the objective is optimized at the token level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip}(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) - \beta D_{\text{KL}}(\pi_{\theta} || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_{\theta}(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage for every token $y_{i,t}$ in a response y_i is set to the same sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Crucially, every token within a given response y_i inherits one and the same advantage $\hat{A}_i$, a quantity derived from the sequence-level reward.

Agentic Search as a Markov Decision Process. We conceptualize the iterative process of agentic search and reasoning as a sequential decision-making problem, formalized as a discrete-time, finite-horizon Markov Decision Process (MDP).

- **State (s_t):** At turn t , the state s_t encapsulates the entire interaction history: the initial question q , the totality of previously generated thoughts and search queries, and the retrieved evidence returned by the environment. Because search results must be interpreted and subsequently integrated into ensuing decisions, the state grows ineluctably with the trajectory, thereby engendering long-horizon dependencies that conventional token-level RL struggles to optimize.
- **Action (a_t):** An action constitutes a complete textual segment generated by the policy π_{θ} , comprising free-form reasoning followed by a decision. The action terminates either with a `</tool_call>` token, which triggers a search and situates the results within `</tool_response>`, or with a `</answer>` token, which brings the trajectory to a close. This structured action space compels the agent to learn not merely *what* to generate but equally *when* to search—a desideratum that introduces considerable variability in trajectory length.
- **Policy (π_{θ}):** The policy π_{θ} is the underlying LLM, which generates tokens autoregressively in conditioning upon the state. Although the policy conditions on the full history, the optimization target is computed solely over the model-generated thoughts and actions, with the retrieved content returned by the environment masked out (Jin et al., 2025b). Optimizing this policy demands credit assignment across long sequences in which numerous intermediate reasoning steps receive neither direct supervision nor reward, a circumstance that further accentuates the necessity of stable sequence-level optimization.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ chronicles all reasoning and tool interactions undertaken for a given question, encompassing both model-generated actions and environment-returned search results. Because the final reward is conferred at the level of the trajectory in its entirety, the optimization problem is fundamentally sequence-level: every early decision may reverberate in the eventual correctness of the answer.
- **Reward ($R(\tau)$):** We adopt a sparse terminal reward: a trajectory receives $R = 1$ whenever the final answer contains the ground-truth text, and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we explicitly incorporate the search engine, denoted $\mathcal{S}$, as a constituent of the environment. The policy LLM π_{θ} learns to generate trajectories by interleaving reasoning with invocations of $\mathcal{S}$. The overarching optimization problem is to identify a policy that maximizes the expected reward, regularized by a KL divergence term that forestalls large deviations from a reference policy π_{ref} :

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot|x;\mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_{\theta}(y|x;\mathcal{S}) || \pi_{\text{ref}}(y|x;\mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_{\theta}(\cdot|x;\mathcal{S})$ signifies that the trajectory y arises from a multi-step process implicating both the policy’s token generation and the information returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) have successfully cast agentic search as an RL problem, the application of conventional algorithms such as PPO or GRPO encounters formidable obstacles. The open-ended character of the search environment aggravates the instability inherent to token-level optimization. A cardinal difficulty resides in the fundamental mismatch between the unit at which the sequence-level reward is assigned and the unit at which token-level optimization proceeds (Zheng et al., 2025). This discrepancy begets high-variance gradient estimates that accumulate over long trajectories and frequently culminate in policy collapse. Moreover, the sparse binary reward signal entails that many training batches comprise exclusively successful or exclusively unsuccessful trajectories, yielding an anomalous advantage and hence no learning signal whatsoever—an outcome that drastically diminishes sample efficiency (Yu et al., 2025; Liu et al., 2025). DSPO is expressly devised to counteract these two critical failure modes in tandem.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO advances two key innovations over prior methods: (1) it conducts policy optimization at the sequence level, thereby aligning the training objective with the trajectory-based reward structure, and (2) it embeds a dynamic filtering mechanism that ensures every training batch furnishes a high-quality, non-zero learning signal. The entire training process, which integrates these two components, is depicted in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Drawing inspiration from GSPO (Zheng et al., 2025), we supplant the unstable token-level importance ratio with a theoretically grounded sequence-level counterpart. The sequence-level importance ratio $s_i(\theta)$ for a response y_i is defined as the geometric mean of its constituent token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length normalization is indispensable for curtailing variance and for ensuring that $s_i(\theta)$ remains confined to a consistent numerical range irrespective of sequence length—a property that is itself vital to stable clipping.

Gradient Analysis. The gradient analysis that follows elucidates precisely why DSPO enhances stability. The gradient of the token-level GRPO objective (in its unclipped form) rescales each token’s log-probability gradient by a noisy, token-specific weight $r_{i,t}(\theta)$. By contrast, the gradient of our sequence-level objective rescales the averaged log-probability gradient of the entire sequence by a single, appreciably more stable sequence-level weight $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying a single, holistic correction factor to the trajectory in its entirety, DSPO forestalls the accumulation of token-level noise that besets prior methods, and thereby engenders training that is fundamentally more stable. In parallel, the dynamic filtering mechanism guarantees a well-conditioned advantage signal $\hat{A}_i$ by constructing training batches from rollout groups that encompass both successes and failures, thus averting the squandering of samples in sparse-reward environments.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary character of our reward function poses an acute challenge for group-based advantage estimation. Should all G responses in a group prove correct ($R = 1$) or all prove incorrect ($R = 0$), the normalized advantage $\hat{A}_i$ degenerates to zero or becomes undefined. Batches of this kind afford no useful gradient signal and consequently squander computational resources.

To overcome this predicament, DSPO incorporates a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, we retain only those groups of trajectories that exhibit an admixture of successful and unsuccessful

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
```

outcomes. A group $\{y_i\}_{i=1}^G$ is admitted for training if and only if its rewards $\{R_i\}_{i=1}^G$ satisfy:

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This condition ensures that the reward variance within every training group is non-zero, thereby guaranteeing a meaningful advantage signal. Such dynamic selection curates a high-quality dataset for each policy update, effectively transmuting a sparse-reward problem into a dense and efficient learning signal.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

By integrating the foregoing components, we arrive at the final DSPO objective. For each valid group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ by means of group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant for numerical stability. The policy π_{θ} is updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined as the geometric mean of the token-level ratios. Crucially, during the likelihood computation we apply loss masking to all tokens retrieved from the search tool, in keeping with Jin et al. (2025b). This safeguard ensures that the model learns to marshal external knowledge in the service of reasoning rather than merely to reproduce it verbatim. The full training procedure is delineated in Algorithm 1.

4 EXPERIMENTS

In this section, we undertake a series of experiments intended to validate empirically the efficacy of our proposed Dynamic-filter Sequence-level Policy Optimization (DSPO) algorithm. Our principal objectives are to establish that DSPO: (1) attains exceptional performance on challenging question-answering benchmarks; (2) manifests markedly enhanced training stability, thereby eschewing the catastrophic collapse that afflicts baseline methods; and (3) derives its performance gains from the synergistic interplay of its core components rather than from any single one in isolation.

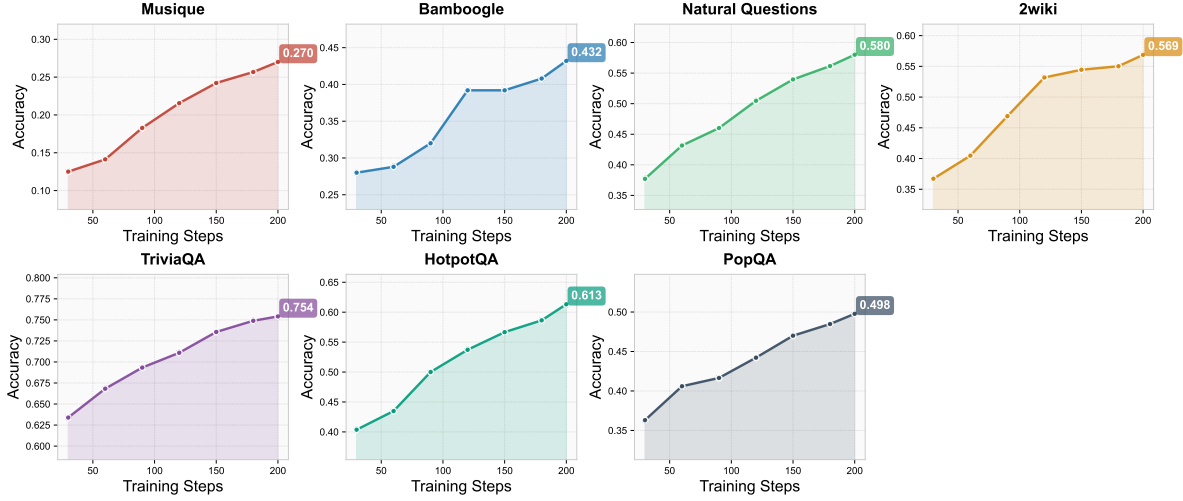


Figure 2: **Validation performance of DSPO across seven benchmarks over the course of training.** The steady, monotonic ascent in accuracy corroborates that DSPO’s reward improvement translates directly into enhanced generalization, and that our method thereby acquires a robust search-and-reasoning policy.

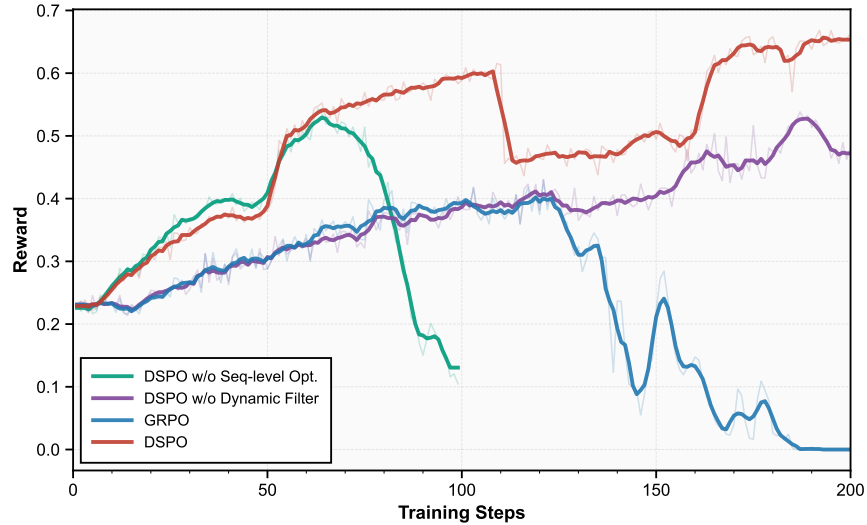


Figure 3: **Training reward dynamics of DSPO and its ablations.** A comparative view of the learning curves. DSPO (red) exhibits stable and monotonic improvement, whereas the token-level variants (green, blue) succumb to catastrophic policy collapse, and the sequence-level variant divested of our filter (purple) plateaus at a conspicuously suboptimal level.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), and as displayed in Table 1, we employ a prompt template to govern the model’s actions throughout the search task, encompassing the `<think>`, `<tool_call>`, and `<answer>` directives.

Benchmarks and Baselines. To afford a rigorous evaluation, our experimental design conforms to the established protocol of Search-R1 (Jin et al., 2025b). We train our model on a composite dataset assembled from the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), and thereafter appraise its generalization capabilities on the test sets of seven diverse QA benchmarks: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: The prompt template employed throughout our experiments.

Our comparison suite embraces both strong external baselines and critical internal ablations. The external baselines comprise the Qwen2.5-7B and 14B models trained with PPO and GRPO within the Search-R1 framework (Jin et al., 2025b;a). In order to deconstruct our method, we further introduce two internal baselines as ablations: (1) **DSPO w/o dynamic filter**, which is tantamount to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to a strong token-level policy, namely DAPO (Yu et al., 2025).

With respect to implementation, we adopted the Qwen2.5-7B-Instruct model as the initial checkpoint for the entirety of our training. Our experiments are erected upon the VeRL framework (Sheng et al., 2025), for which we adapted the provided Search-R1-like example code and scripts so as to accommodate our methodology. We benchmark DSPO against a comprehensive suite of baselines: for external comparison we invoke the PPO and GRPO methods of the Search-R1 framework (Jin et al., 2025b;a). Crucially, inasmuch as our work employs a modified reward function, we retrained these models under precisely our own experimental conditions so as to ensure an equitable comparison, and the results of these retrained models constitute our primary external benchmarks.

Implementation and Evaluation. In order to isolate the benefits attributable to our algorithm, all RL experiments deliberately enlist a standard BM25 retriever. This controlled configuration ensures that the observed performance improvements are ascribable directly to the model’s learned policy rather than to any sophistication of the retriever. Across all methods, models are trained using a sparse, binary reward signal predicated on the substring Exact Match (subEM) of the final answer, and subEM likewise serves as the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

In order to convey a holistic view of our algorithm’s efficacy, we present a comprehensive comparison in Table 2. Given the synergistic nature of DSPO’s components, we deem it most illuminating to present our main results in concert with our ablation study. This single table juxtaposes DSPO against both external state-of-the-art baselines and its own ablated variants, thereby affording a lucid and direct assessment of its overall superiority as well as of the indispensability of each of its core components.

Table 2: A comprehensive comparison of DSPO with baselines and ablation variants across seven QA benchmarks. The baselines comprise the Search-R1 models (7B & 14B) trained with GRPO and PPO (Jin et al., 2025b;a), while the ablations excise key components, namely ‘w/o dynamic filter’ and ‘w/o seq-level opt.’. The original EM scores reported by Search-R1 are given in parentheses; to preserve consistency of evaluation, we retrained and re-evaluated these models under our adjusted rewards. Best results are rendered in **bold** and second-best are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results marshalled in Table 2 underscore DSPO’s unequivocal superiority. Our DSPO-trained 7B agent attains a remarkable average score of 0.531, thereby establishing a new state of the art—a figure that corresponds to a 34.1% relative improvement over the identically sized Search-R1 (GRPO, 7B) model. More striking still, our 7B agent secures a marginally superior average score to that of the considerably larger Search-R1 14B models (under both GRPO and PPO). This outcome furnishes compelling evidence that the performance gains issue from a more effective and stable learning algorithm rather than from any over-reliance upon model scale.

Analysis of Ablations. The ablation results, likewise presented in Table 2, demonstrate unequivocally that both of DSPO’s components are indispensable. First, excising the dynamic filter (‘w/o dynamic filter’) precipitates a catastrophic deterioration in performance, with the average score plummeting to 0.313. This decline throws its critical role into relief: bereft of the filter, the sequence-level objective is starved of any useful learning signal on account of homogeneous-reward batches. Second, ablating sequence-level optimization (‘w/o sequence-level opt.’) likewise occasions a pronounced degradation, yielding an average score of 0.406. Although this token-level variant surpasses its filter-less counterpart, it falls conspicuously short of the full DSPO model and, as we demonstrate in the ensuing section, remains susceptible to catastrophic training instability. These observations confirm that the synergy is decisive: sequence-level updates are essential to stability, whereas our dynamic filter is essential to transmuting sparse rewards into an efficient learning signal.

Beyond these quantitative metrics, we observe that DSPO elicits sophisticated search behaviors—among them the recognition of irrelevant results, query reformulation, and multi-turn verification (see Appendix A for detailed trajectory examples). All such behaviors emerge from pure RL training conducted through DSPO.

4.3 ANALYSIS OF TRAINING DYNAMICS

In order to validate empirically our claims concerning stability and efficiency, we analyze the training reward dynamics of DSPO, its ablations, and the key baselines. Figure 3 offers a compelling visualization of these dynamics that reinforces our core architectural choices.

DSPO (red) exhibits a smooth, monotonic ascent and converges efficiently to the highest reward level, a trajectory that empirically corroborates the stability conferred by its sequence-level objective. In stark contradistinction, the **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—succumb to catastrophic policy collapse early in training: their rewards plummet in the aftermath of a fleeting initial improvement, a patent manifestation of the instability wrought by high-variance, token-level gradient updates. Meanwhile, **DSPO w/o Dynamic Filter** (purple), which capitalizes upon sequence-level updates yet is deprived of an efficient learning signal, remains stable but plateaus at a conspicuously suboptimal performance ceiling. These dynamics reveal that DSPO’s synergy of sequence-level stability and dynamic filtering is the linchpin of its robust and effective policy optimization.

To ascertain that these improvements in training reward betoken genuine generalization rather than reward hacking, we monitor validation performance on the key benchmarks throughout training. As illustrated in Figure 2, DSPO’s validation accuracy on NQ, HotpotQA, and the remaining diverse benchmarks rises consistently, mirroring its stable reward curve. This correlation confirms that the agent is indeed acquiring a generalizable search-and-reasoning policy.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To validate the robustness of our approach yet further, we extend our evaluation so as to interrogate both model scalability and domain generalization.

Scalability to Larger Models. We investigate whether the stability benefits of DSPO carry over to larger parameter scales by training Qwen2.5-14B-Instruct. As detailed in Table 3, DSPO evinces remarkable scalability: the DSPO-trained 14B model attains an average accuracy of 60.6%, thereby outstripping the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%**. These results confirm that our method exploits the augmented model capacity to good effect, yielding a performance that consistently exceeds the standard baselines.

Generalization to Mathematical Reasoning. We further gauge the universality of DSPO by applying it to single-turn mathematical reasoning tasks drawn from the Qwen2.5 and Qwen3 model families. Table 4 sets forth the comparison on Math500 and Olympiad-Bench, wherein DSPO consistently surpasses GRPO across both the 7B and the 4B model sizes. This corroborates that DSPO remains effective across general reasoning domains beyond agentic search.

Table 3: A scalability analysis conducted on Qwen2.5-14B-Instruct. Best results are rendered in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Table 4: Generalization to the domain of mathematical reasoning. Best results are rendered in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

In this work, we grappled with the critical instability and sample-inefficiency afflictions that beset RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), a refined algorithm that secures robust training by way of two key components: sequence-level optimization, which forestalls catastrophic policy collapse, and a dynamic outcome-based filter, which transmutes sparse rewards into a consistently effective learning signal. Our experiments demonstrated that DSPO not only attains substantial performance across a suite of challenging question-answering benchmarks but likewise exhibits training stability superior to that of prior methods.

By enabling robust training from environmental feedback alone, DSPO establishes a practical and efficient blueprint for the construction of capable LLM agents unencumbered by costly expert data. Upon this stable foundation, future work may confidently pursue the integration of advanced retrievers or the extension of DSPO to complex, multi-tool settings. Furthermore, inasmuch as the challenges of sparse rewards and unstable policy gradients are by no means peculiar to search, we conjecture that the principles underlying DSPO will confer comparable gains in performance and stability upon other domains such as mathematics and code generation—a promising avenue that awaits future validation. We are persuaded that the core tenets of DSPO—the reconciliation of the optimization unit with the reward signal, together with the guarantee of signal density—will prove instrumental in the development of the next generation of autonomous AI.

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A APPENDIX

In order to afford insight into the learned search strategies, Table A.1 sets forth representative trajectories generated by our model. These examples attest to the model’s capacity to: (1) reformulate its queries whenever the initial searches surrender irrelevant results, (2) corroborate information by means of follow-up searches, and (3) distill comprehensive answers from the retrieved results.

Table A.1: Representative trajectories that exemplify our model’s learned search and reasoning capabilities. The model manifests sophisticated behaviors, encompassing **(a)** the recognition of irrelevant results, **(b)** query reformulation, and **(c)** verification strategies.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let’s try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin’s occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American journalist, attorney, soldier, and politician. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician